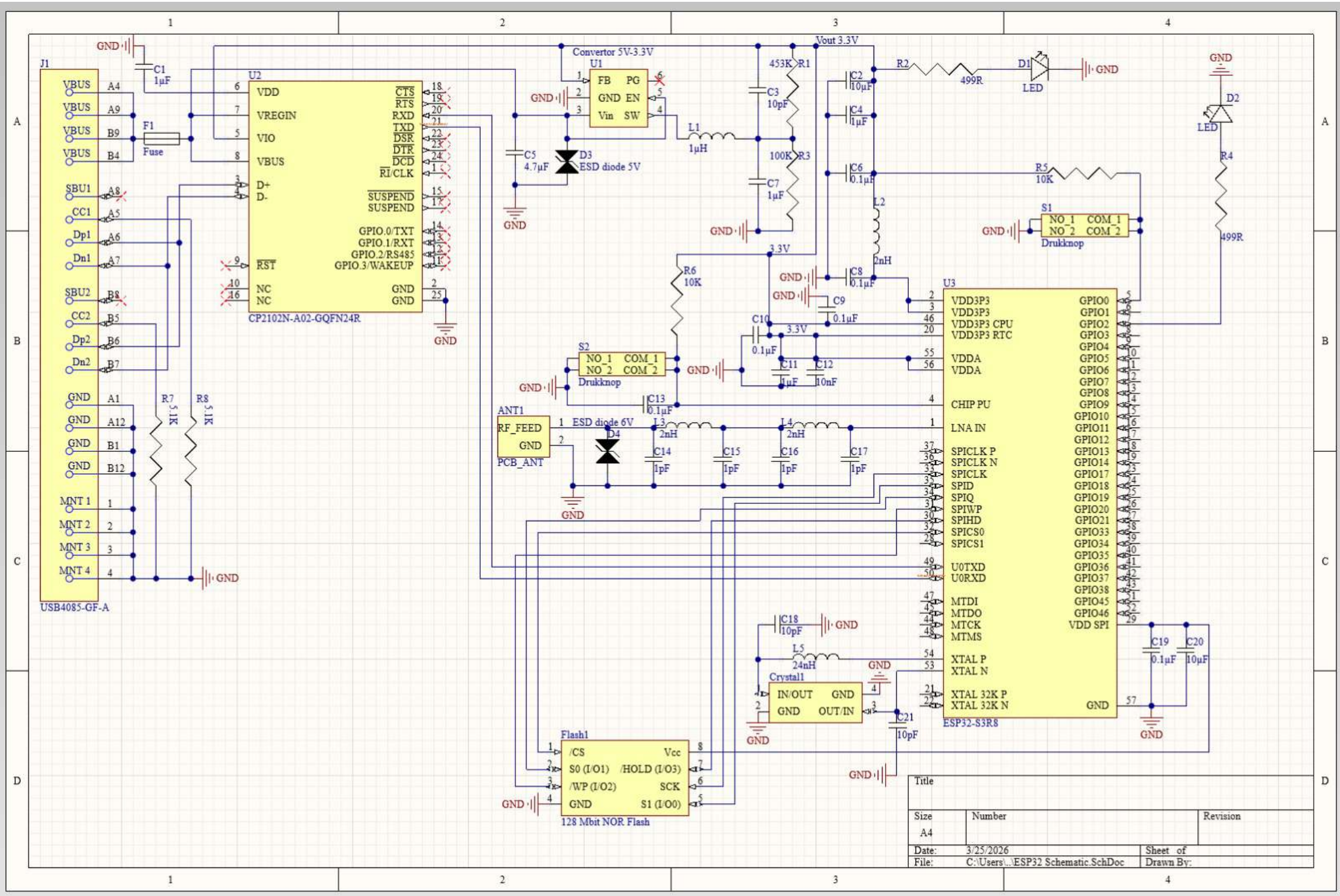
J1
USB4085-GF-A
VBUS A4
VBUS A9
VBUS B9
VBUS B4
SBU1 A8
CC1 A5
Dp1 A6
Dn1 A7
SBU2 B8
CC2 B5
Dp2 B6
Dn2 B7
GND A1
GND A12
GND B1
GND B12
MNT 1 1
MNT 2 2
MNT 3 3
MNT 4 4
C1 1µF
F1 Fuse
R7 5.1K
R8 5.1K
U2
CP2102N-A02-GQFN24R
VDD
VREGIN
VIO
VBUS
D+
D-
RST
NC
NC
CTS
RTS
RXD
TXD
DSR
DTR
DCD
RI/CLK
SUSPEND
SUSPEND
GPIO.0/TXT
GPIO.1/RXT
GPIO.2/RS485
GPIO.3/WAKEUP
GND
GND
Convertor 5V-3.3V
U1
FB PG
GND EN
Vin SW
C5 4.7µF
D3 ESD diode 5V
L1 1µH
Vout 3.3V
453K R1
100K R3
C3 10pF
C7 1µF
C2 10µF
C4 1µF
C6 0.1µF
L2 2nH
C8 0.1µF
3.3V
R2 499R
D1 LED
R5 10K
S1
NO_1 COM_1
NO_2 COM_2
Drukknop
D2 LED
R4 499R
R6 10K
C9 0.1µF
C10 0.1µF
C11 1µF
C12 10nF
S2
NO_1 COM_1
NO_2 COM_2
Drukknop
C13 0.1µF
ANT1
RF_FEED
GND
PCB_ANT
D4 ESD diode 6V
L3 2nH
L4 2nH
C14 1pF
C15 1pF
C16 1pF
C17 1pF
U3
ESP32-S3R8
VDD3P3
VDD3P3
VDD3P3 CPU
VDD3P3 RTC
VDDA
VDDA
CHIP PU
LNA IN
SPICLK P
SPICLK N
SPICLK
SPID
SPIQ
SPIWP
SPIHD
SPICS0
SPICS1
U0TXD
U0RXD
MTDI
MTDO
MTCK
MTMS
XTAL P
XTAL N
XTAL 32K P
XTAL 32K N
GND
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO33
GPIO34
GPIO35
GPIO36
GPIO37
GPIO38
GPIO45
GPIO46
VDD SPI
C18 10pF
L5 24nH
Crystal1
IN/OUT GND
GND OUT/IN
C21 10pF
C19 0.1µF
C20 10µF
Flash1
/CS
S0 (I/O1)
/WP (I/O2)
GND
Vcc
/HOLD (I/O3)
SCK
S1 (I/O0)
128 Mbit NOR Flash
Title
Size A4
Number
Revision
Date: 3/25/2026
Sheet of
File: C:\Users\..\ESP32 Schematic.SchDoc
Drawn By: